additional trace materials → such as →

Fe^{2+} (iron) on avg less than 20% of Fe consumed is absorbed → upon absorption ~~Heme~~ → Fe becomes part of the total pool.

Bone Marrow, liver, Spleen → store ~~it~~ Fe in protein compounds ferritin (1) & hemosiderin (2)

ferritin → & ~~hemosiderin~~

→ transports the Fe across the intestinal cell plasma membranes & from its storage sites into tissue fluid where it enters the blood.

Ferroportin

When EPO stimulates the production of RBCs Fe is released from storage, bound to transferrin & carried to the red marrow → production of erythrocytes, iron is released from storage → bound to tranferrin & carried to red marrow

Cu → a trace element/mineral
is a component of two plasma proteins
hephaestin & ceruloplasmin → without these hemoglobin could not be ~~adqeo~~ adequatey produced.

{located in the intestinal ~~ti~~ villi}

hephaestin → enables Fe^{2+} to become Fe^{3+}